Sooraj Sunil
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Contents

Education

Doctor of Philosophy, Electrical and Computer Engineering Sept 2023 – Present
University of Windsor, ON

Master of Applied Science, Electrical and Computer Engineering May 2019 – Dec 2021
University of Windsor, ON

Bachelor of Engineering, Electronics and Communication Engineering Aug 2014 – Apr 2018
Anna University, India

Work Experience

Doctoral Teaching and Research Assistant, BMS Lab Sept 2023 – Present
University of Windsor, ON

- Ph.D. research centered on Li-ion battery modeling, state estimation, and control strategies to enhance the safety, performance, and reliability of energy storage systems.

Full-time Research Assistant, BMS Lab Aug 2022 – Aug 2023
University of Windsor, ON

- Led battery data planning, collection, and analysis in MATLAB to advance battery SOx estimation algorithms.
- Worked with a multidisciplinary research team to prototype a wireless battery management system.

Master's Teaching and Research Assistant, Mechatronics Lab May 2019 – Dec 2021
University of Windsor, ON

- Master's research centered on design and development of novel feature-based occupancy map-merging algorithm to improve autonomous multi-robot navigation.

Recognition & Scholarships

Entrance Scholarship (CDN $7500) May 2019
University of Windsor, Canada

Professional Memberships & Certifications

Student Member Sept 2023 – Present
Institute of Electrical and Electronics Engineers (IEEE)

Lithium Battery Safety Oct 2023
Lion Technology Inc.

Mentoring

Graduate Mentor, University of Windsor Formula Electric Team Jan 2023 – Present
University of Windsor, ON

- A humble part of an incredible journey — from battery module design (2023) to a fully functional EV battery pack (2025).
- Mentored a multidisciplinary undergraduate engineering team in the design, construction, and testing of a high-performance EV battery pack and battery management system.

Graduate Mentor, University of Windsor Capstone Teams Jan 2023 – Present
University of Windsor, ON

- FSAE Accumulator Design (2025) — Ali Elshikh, Zuhair Abdo, Sahana Jayatilaka, Jared Brouerius Van Nidek
- SOH Indicator for a Battery Fuel Gauge (2024) — Sam Letwin, Nicolas Manzon, Andy Garro
- Power System Modeling for the Formula Electric Team (2023) — Saad Arshad, Shaan Batra, Aleks Dejanovic, Luka Tojcic
- Ionic Wind Pump for Battery Cooling (2023) — Sabreen Elastal, Asra Kala, Simrat Poonia, Matthew Aguilar, Boron Al-Diraie
- Active Cell Balancing for an EV Battery Pack (2023) — Justin Haddad, Luca Mannarino, Nick Nguyen

Synergistic Activities

Conference Judge, UWill Discover Student Research Conference Mar 2026
University of Windsor, ON

Peer Reviewer (7 Reviews for 4 Journals) 2023 – Present
Journal of Energy Storage, Journal of Power Sources, Measurement, MethodsX

Publications

Journals:

[J5] Sunil, S., Sundaresan, S., Pattipati, K.R., and Balasingam, B. (Under Review). The Role of Open-Circuit Voltage Characterization in Model-Based Capacity Estimation. *IEEE Transactions on Instrumentation and Measurement.*

[J4] Sunil, S., Pillai, P., Pattipati, K.R., and Balasingam, B. (2025). Two-Stage Least Squares for Equivalent-Circuit Parameter Estimation of Lithium-Ion Batteries Using Pulse-Relaxation Excitation. *IEEE Transactions on Instrumentation and Measurement.*

[J3] Pillai, P., Desai, S., Sunil, S., Pattipati, K.R., and Balasingam, B. (2025). Least Squares Approach to Improve Standardized Estimation of OCV and Internal Resistance in Batteries. *IEEE Journal of Emerging and Selected Topics in Industrial Electronics.*

[J2] Sunil, S., Pattipati, K.R., and Balasingam, B. (2025). Piecewise Linear Approximation of Battery Open-Circuit Voltage Characteristics Using Dynamic Programming. *IEEE Transactions on Instrumentation and Measurement.*

[J1] Sunil, S., Mozaffari, S., Singh, R., Shahrrava, B., and Alirezaee, S. (2023). Feature-Based Occupancy Map-Merging for Collaborative SLAM. *Sensors*, 23(6), 3114.

Conferences:

[C10] Sunil, S., and Balasingam, B. (Accepted). Deep Learning vs. Model-Assisted Learning: Objective Performance Evaluation of the LSTM Classifier Using Time Series Data. In *2026 29th International Conference on Information Fusion.*

[C9] Sunil, S., Pattipati, K.R., and Balasingam, B. Improved Linear Estimation of Battery Open-Circuit Voltage and RC Model Parameters from Pulse-Relaxation Data. In *2025 IEEE 4th Industrial Electronics Society Annual On-Line Conference (ONCON)* (pp. 1–6). IEEE.

[C8] Nguyen, N., Sunil, S., Pillai, P., and Balasingam, B. An Offsetting-Based Correction to Improve the Accuracy of Low-Rate OCV Curves for Lithium-Ion Batteries. In *2025 IEEE 34th International Symposium on Industrial Electronics (ISIE)* (pp. 1–6). IEEE.

[C7] Mahmoodzadeh, M., Sunil, S., Pillai, P., Biondi, F., Strayer, D.L., Cooper, J.M., McDonnell, A.S., and Balasingam, B. Temporal Analysis of Cognitive Workload During Manual and Partial Driving Automation Based on the Detection Response Task. In *2025 IEEE 34th International Symposium on Industrial Electronics (ISIE)* (pp. 1–5). IEEE.

[C6] Sundaresan, S., Sunil, S., Balasingam, B., and Pattipati, K.R. Joint Estimation of Open-Circuit Voltage and Equivalent Circuit Model Parameters Using State-Space Model Optimization. In *2023 IEEE Transportation Electrification Conference & Expo (ITEC)* (pp. 1–6). IEEE.

[C5] Sunil, S., Sundaresan, S., Pillai, P., and Balasingam, B. Inverse Characterization of Open-Circuit Voltage for State-of-Charge Estimation of Batteries. In *2023 IEEE Transportation Electrification Conference & Expo (ITEC)* (pp. 1–6). IEEE.

[C4] Sunil, S., Balasingam, B., and Pattipati, K.R. State-of-Charge Estimation of Batteries Using the Extended Kalman Filter: Insights into Performance Analysis and Filter Tuning. In *2022 IEEE 1st Industrial Electronics Society Annual On-Line Conference (ONCON)* (pp. 1–6). IEEE.

[C3] Sunil, S., Mozaffari, S., Singh, R., Shahrrava, B., and Alirezaee, S. LiDAR-Based Cooperative SLAM with Different Parameters. In *2022 7th International Conference on Mechanical Engineering and Robotics Research (ICMERR)* (pp. 82–87). IEEE.

[C2] Sundaresan, S., Sunil, S., Balasingam, B., and Pattipati, K.R. Fast Offline Battery Capacity Estimation Approach with Performance Bounds. In *2022 IEEE Electrical Power and Energy Conference (EPEC)* (pp. 188–193). IEEE.

[C1] Sunil, S., Sundaresan, S., Balasingam, B., and Pattipati, K.R. Novel Table-Based Kalman Filter for State-of-Charge Estimation of Batteries. In *2022 IEEE Electrical Power and Energy Conference (EPEC)* (pp. 200–205). IEEE.

Thesis:

[T1] S. Sunil. "Heterogeneous Collaborative Mapping for Autonomous Mobile Systems", *University of Windsor*, Canada, Jan. 2022.